c) $f: Z \times Z \to \mathbb{Q}$, $f(m,n) = m/n$.

Soln: If $n=0$, then $f(m,n) = m/0$ which is undefined.

Domain: $Z \times Z$ Codomain: $\mathbb{Q}$ Domain of defn: $Z \times (Z - \{0\})$

Set of values for which f is not defined: $\{(m,0)\}$, where $m \in Z$.

Not a total function. □

d) $f: Z \times Z \to Z$, $f(m,n) = mn$

Soln: Domain: $Z \times Z$ Co-domain: Z

Domain of Defn: $Z \times Z$, $\because mn$ is defined for all values in domain

Set of values where f is not defined: ϕ. Total function □

e) $f: Z \times Z \to Z$, $f(m,n) = m - n$ if $m > n$.

Soln: Domain: $Z \times Z$ Co-domain: Z.

Domain of Defn: (x,y) where $x \in Z$ and $y \in Z$, $x > y$.

Set of values where f is not defined: $\{(m,n) \mid m \not> n,\ m,n \in Z\}$

Not a total function. □

8) a) Show that a partial function from A to B can be viewed as a function f^* from A to $B \cup \{u\}$, where u is not an element of B and

$$f^*(a) = \begin{cases} f(a) & \text{if } a \text{ belongs to the domain of definition of } f. \\ u & \text{if } f \text{ is undefined at } a. \end{cases}$$

Soln: $\because f$ is a partial function, $\exists x \in A$ s.t. $f(x)$ is not defined, ie. $x \notin$ Domain of defn. of A. $\therefore f(x) \notin B$.

By given defn. of f^*, $f^*(x) = u \notin B$.

If $a \in$ domain of definition of f, $f^*(a) = f(a)$.

Domain of f^*: A Co-domain of f^*: $B \cup \{u\}$.

It's easy to see that the Domain of Defn. of f^*: A.

$\therefore$ The set of values for which f^* is undefined is ϕ

So, f^* is a total function. □